GG-W-239C
July 19, 1985
SUPERSEDING
GG-W-239B
May 13, 1976

FEDERAL SPECIFICATION

WHEELS, ABRASIVE, (Silicon Carbide, Mounted, Dental)

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers multi-shape silicon carbide abrasive wheels (mounted points) for use with dental handpieces.

1.2 Classification.

1.2.1 Types and shapes. Mounted abrasive wheels shall be of the following types and shapes, as specified (see 6.1):

Type I - For angle handpiece.
  Shape D - Cone.

Type II - For straight handpiece.
  Shape A - Rounded edge.
  Shape B - Square edge.
  Shape C - Knife edge.
  Shape D - Cone.
  Shape E - Large rounded cone.
  Shape F - Inverted cone.
  Shape G - Tapered cylinder.
  Shape H - Cylinder.
  Shape I - Ball.

2. APPLICABLE DOCUMENTS

2.1 The issues of the following documents, in effect on date of invitation for bids or solicitation for offers, form a part of this specification to the extent specified herein.

FSC 6520

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Federal Specifications:

GG-I-526 - Instruments, Dental and Surgical, General Specifications.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-601 - Boxes, Wood, Cleated-plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Setup.

Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards, and Commercial Item Descriptions. The Index, which includes cumulative bi-monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification, other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Washington, DC; Philadelphia, PA; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specification:

MIL-P-116 - Preservation, Methods of
MIL-L-10547 - Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment uand Storage.
MIL-STD-174 - Palletized Unit Loads.
MIL-STD-794 - Parts and Equipment, Procedures for Packaging of.

(Copies of military specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards Institute (ANSI) Standard:

B 74.12 - Size of Abrasive Grain - Grinding Wheels, Polishing and General Industrial Uses.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10018).

American Society for Testing and Materials (ASTM) Standards:

E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials, Tests for.

E 92 - Vickers Hardness of Metallic Materials, Tests for.

(Application for copies should be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification, shall take precedence.

3. REQUIREMENTS

3.1 Material.

3.1.1 Abrasive. The abrasive shall be silicon carbide, crystalline in form. The abrasive shall be clean and shall be free from all extraneous material.

3.1.2 Bond. A vitrified bond shall be used.

3.1.3 Mandrels. Mandrels shall be made of either corrosion-resisting steel or carbon steel suitable for the purpose intended. Mandrels made of carbon steel shall be plated or coated to render corrosion resistant.

3.2 Style, design and dimensions. Wheels (mounted points) shall be mounted true on the mandrels and shall be permanently attached. The portion of the mandrel inserted in the wheel shall be sufficiently knurled or shaped in such a manner as to eliminate any possibility of pulling off under normal operating conditions. The mandrel shall have sufficient strength to prevent breaking of the neck under normal use. Wheels shall be suitable for use with dental handpieces, specified for abrading or cutting away tooth

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structure, artificial teeth, and dental restorations. Type I and Type II, wheels shall be mounted on a mandrel fabricated of material as specified in 3.1.3. Mandrel shall conform to design and dimensions as shown in figure 1, for the applicable type specified, and shall dimensionally interface with the applicable handpiece. Abrasive wheels shall be of the shapes specified, and shall conform to the dimensions shown in figure 2. The overall length of the mounted abrasive wheels (mandrel plus head) shall be as shown in table I for the type and shape specified (see 1.2.1 and 6.1).

TABLE I. Overall length of mounted abrasive wheels.

Type of shank	Shape of head	Overall length of mounted abrasive wheels (Mandrel plus head), inches
I	D	15/16 ± 1/32
II	C	1-11/16 ± 3/32
II	A, F, I	1-3/4 ± 3/32
II	B, D, H	1-13/16 ± 3/32
II	E, G	1-7/8 ± 3/32

3.2.1 Figures. The figures forming a part of this specification are specification control figures. The supplier shall be responsible for preparing his own shop drawings. Where tolerances presribed could cumulatively result in improper design, the supplier shall provide a product with dimensions within the tolerances prescribed on the drawing to insure proper design, function, fit, assembly, or operation of the item. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer.

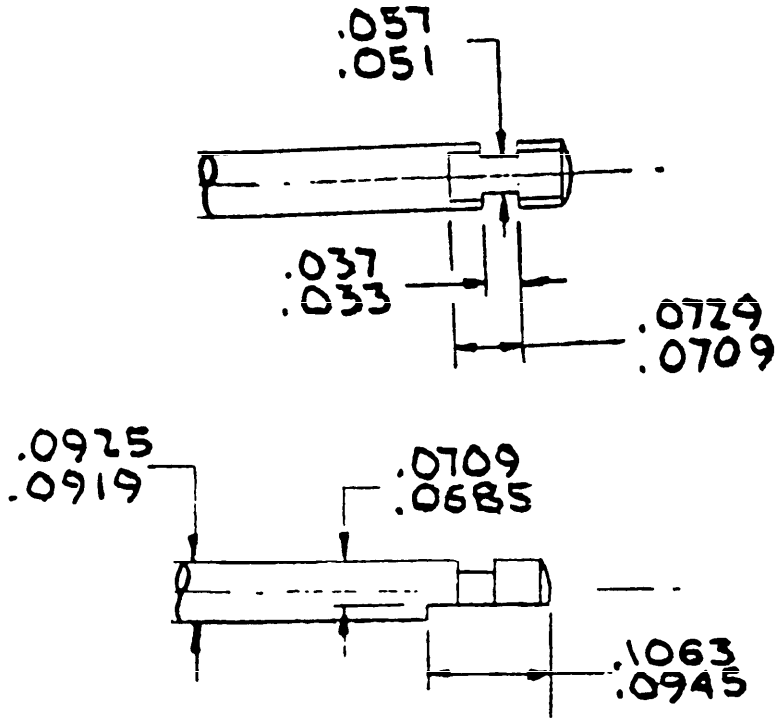
3.3 Abrasive size. Abrasive shall be 120 grit for shapes A, B, D, E, G, H and I, and abrasive shall be 150 grit for shapes C and F, when tested as specifed in 4.4.1.

3.4 Hardness. Rockwell hardness of the mandrels shall be C20 minimum, when tested as specified in 4.4.2.

3.5 Boil test. The mandrels shall show no signs of corrosion when tested as specified in 4.4.3.

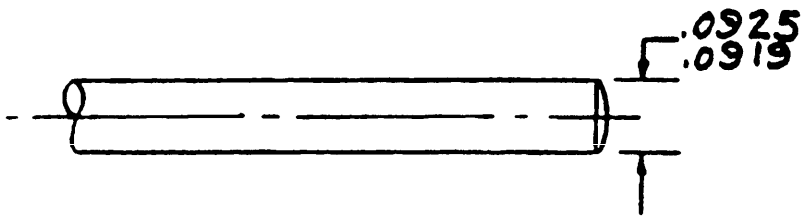
3.6 Grinding characteristics. When tested as specified in 4.4.4, the wheels, except for shape C, shall cut a groove to a depth of one-third or more of the largest diameter of the wheel in a piece of plate glass. At the end of the test period, the wheel shall not be worn more than 15 percent of its original diameter. Shape C wheels shall cut a groove to a depth of 10 percent or more of the original diameter of the wheel, and shall not be worn more than 5 percent of the original diameter.

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TYPE I

FOR ANGLE HANDPIECE



TYPE II

FOR STRAIGHT HANDPIECE

NOTE: DIMENSIONS ARE IN INCHES

FIGURE 1

DIMENSIONS OF MANDRELS

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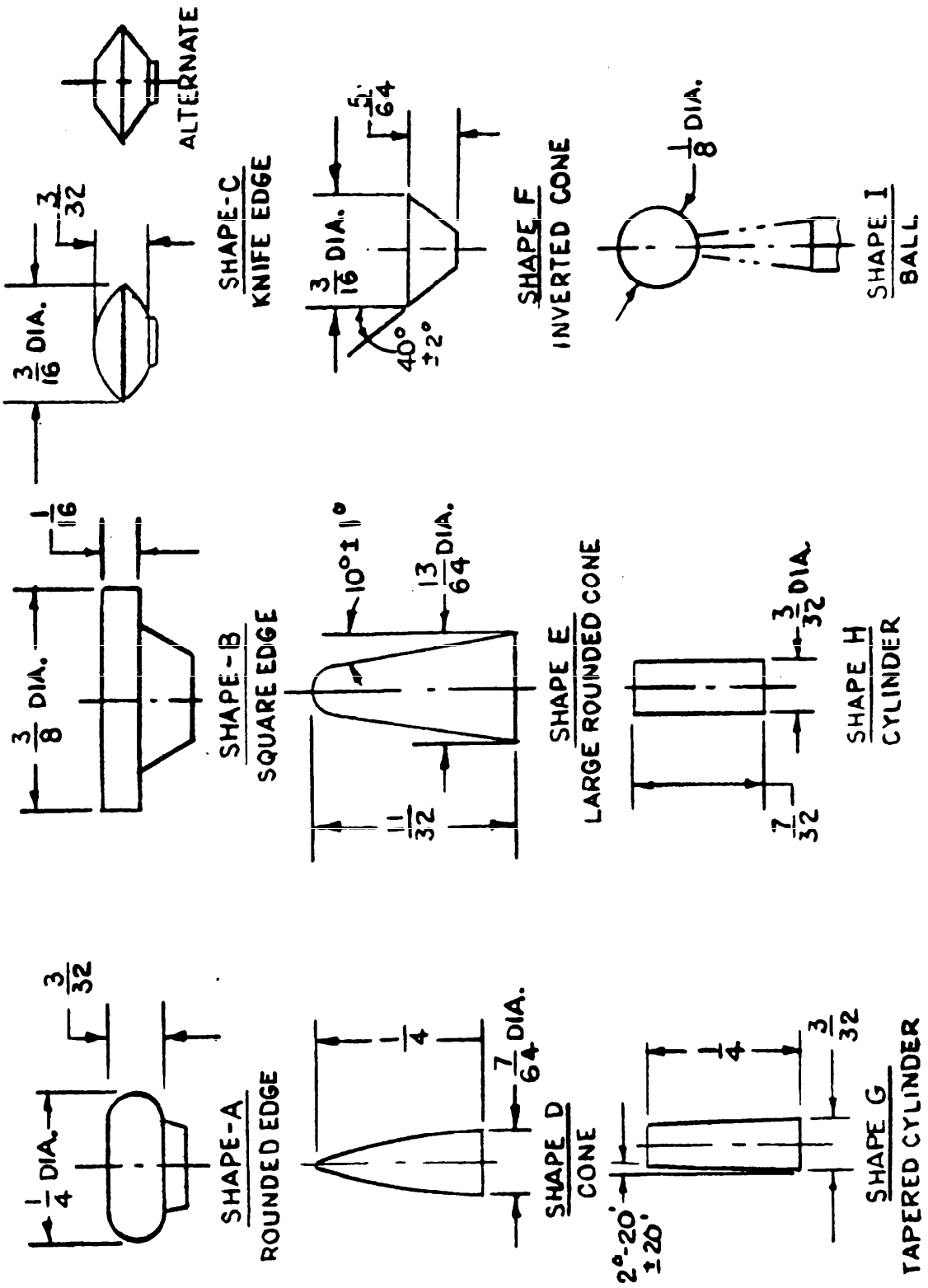


FIGURE 2

SHAPES AND DIMENSIONS OF ABRASIVE WHEELS

NOTE: DIMENSIONS ARE IN INCHES
TOLERANCE ON FRACTIONS: ±5%

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3.7 Runout. When tested as specified in 4.4.5, the total indicated runout for abrasive wheels shall not exceed the following:

Wheels with a head diameter 0.125 inch and under	0.003 inch.
Wheels with a head diameter greater than 0.125 inch and including 0.250 inch	0.004 inch.
Wheels with a head diameter greater than 0.250 inch	0.005 inch.

3.8 Finish. The shanks of the mandrels shall be smoothly finished.

3.9 Identification marking. Marking of the mandrel with the name or registered trademark of the manufacturer is optional.

3.10 Workmanship. The wheels shall be symmetrically shaped and smoothly finished. Wheels shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 year after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certification. The material (abrasive and steel) may be accepted on the basis of a supplier's certification for the abrasive and steel specified.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered are classified in tables II, III, and IV (see 4.2.1, 4.2.2, and 4.3) and 4.5.

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4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQLs) shall be as indicated in table II. Unit of product for examination purposes shall be one wheel.

TABLE II. Sampling for Examination.

Type Defect	Inspection Level	AQL (Percent Defective)
Major	II	1.0
Minor A	II	2.5
Minor B	S-2	4.0

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as specified in table III. The acceptance number shall be zero for all sample sizes. The unit of product for test purposes shall be one wheel.

TABLE III. Test criteria.

Characteristic	Requirement	Test procedure	Inspection level
Abrasive size	3.3	4.4.1	(see 4.2.3)
Hardness	3.4	4.4.2	S-1
Boil test	3.5	4.4.3	S-1
Grinding characteristics	3.6	4.4.4	S-2
Runout	3.7	4.4.5	S-2

4.2.3 Sampling for abrasive particle grading. A sample of 100.0 ± 5.0 grams from each lot or batch of abrasive particles shall be selected for grading.

4.2.4 Sample unit usage. Where feasible, the same unit of product shall be used for the determination of two or more test characteristics.

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4.3 Examination. The abrasive wheels shall be examined to determine compliance with the requirements of this specification. Examination shall be conducted in accordance with the following classification of defects. Examination shall include, but shall not be limited to, the defects listed in table IV.

TABLE IV. Classification of defects.

Categories	Defects
Major:	
101	Type not as specified.
102	Shape not as specified.
103	Diameter Type I mandrel not as specified.
104	Diameter Type II mandrel not as specified.
105	Type I mandrel latch dimensions not as specified.
106	Abrasive wheel split, cracked, chipped or malformed.
107	Abrasive wheel loose on mandrel.
Minor A:	
201	Mandrel shank not smoothly finished.
202	Extraneous materials present in abrasive.
Minor B:	
251	Overall length Type I mounted abrasive wheel not as specified.
252	Overall length Type II mounted abrasive wheel not as specified.
253	Abrasive head dimensions not as specified.

4.4 Test procedures.

4.4.1 Abrasive size. The size of the silicon carbide abrasive grit shall be determined in accordance with ANSI Standard B 74.12.

4.4.2 Hardness. Hardness shall be determined in accordance with ASTM Designation E 18 (Rockwell) or E 92 (Vickers or Diamond Pyramid).

4.4.3 Boil test. The boil test shall be conducted in accordance with GG-I-526.

4.4.4 Grinding test.

4.4.4.1 Glass specimen. The test material shall be plate glass, 0.50 inch thick, having a Knoop hardness number of 450 to 510, with a one kilogram load. The glass specimen shall be of sufficient size to be held securely in the specimen holder.

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4.4.4.2 Method of test. Two sample wheels (except shape C) shall be mounted in a suitable chuck attached to a true running spindle, running at approximately 3600 revolutions per minute (r.p.m.). The plate glass specimen shall be mounted in a holder in such manner that the wheel shall be positioned against the face of the glass, ultimately enabling the full extent of the cutting surface to be engaged in the test. The mandrel shall be parallel to the face of the glass. The face of the glass shall be applied to the revolving wheels under a load of 2.00 pounds for 5 minutes. Shape C points shall be tested under a load of 0.25 pounds at 3600 r.p.m. for 5 minutes. The diameter of the wheels shall be measured to 0.001 inch and the depth of penetration shall be measured to 0.001 inch using a dial gage. A stream of air shall be directed on the field during the testing to insure freedom from clogging debris.

4.4.5 Runout. The mandrel shall be placed and held in a "V" block or precision chuck, and rotated slowly. A dial indicator sensitive to 0.0002 inch and equipped with a knife-edge foot, or a shadow-graph or other suitable equipment of equal accuracy, shall be employed. If a dial indicator is used, the indicator shall be mounted with the shaft perpendicular to the axis of the mandrel, and so positioned that the foot contacts the mandrel at the point nearest the abrasive wheel. The maximum and minimum readings obtained during one complete 360-degree rotation of the mandrel shall be measured and recorded to the nearest 0.0005 inch. The difference between these two readings shall be taken as the total indicated runout.

4.5 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. Packaging.

5.1 Preservation. Preservation shall be level A or C as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning. Wheels shall be cleaned in accordance with MIL-P-116.

5.1.1.2 Drying. Wheels shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservative application. Preservatives shall not be used.

5.1.1.4 Unit package. Twelve wheels of one type and shape only shall be packaged in accordance with MIL-P-116, method III in a box of appropriate size and design constructed in accordance with PPP-B-566 or PPP-B-676. The box design may be a dispensing type. Closure shall be as specified in the applicable box specification. As an alternate, wheels may be blister packaged in multiple compartments. Each compartment shall be furnished with perforation or tear lines to permit easy removal of each wheel; or packaged in a plastic box of appropriate size and design having a friction closure.

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5.1.1.5 Intermediate package. Twelve units containing wheels of one type and shape only shall be packed in a snug fitting box conforming to PPP-B-566, PPP-B-636, class domestic or PPP-B-676. Closure of the box shall be as specified in the applicable box specification.

5.1.2 Level C. The level C preservation for wheels shall conform to MIL-STD-794 requirements for this level.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.1).

5.2.1 Level A. Twenty-four intermediate packages containing wheels of one type and shape only shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Fiberboard boxes shall be waterproofed as specified in PPP-B-636. Closure and strapping shall be as specified in the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall be as specified in the applicable linen specification.

5.2.2 Level B. Twenty-four intermediate packages containing wheels of one type and shape only shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-601, domestic type; PPP-B-621, class 1, or PPP-B-636, class domestic. Closure shall be as specified in the applicable box specification.

5.2.3 Level C. The level C packing for wheels shall conform to MIL-STD-794 requirements for this level.

5.2.4 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

5.2.4.1 Levels A and B. Wheels, packed as specified in 5.2.1 and 5.2.2, shall be unitized on pallets conforming to MIL-STD-147.

5.2.4.2 Level C. Wheels, packed as specified in 5.2.3, shall be unitized as specified in MIL-STD-794.

5.3 Marking.

5.3.1 Civil agencies. Marking shall be as specified in FED-STD-123.

5.3.2 Military activities.

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5.3.2.1 Levels A, B, and C. Each unit package, intermediate package, exterior container and unitized load shall be marked as specified in MIL-STD-129. In addition, the name of the contractor shall be shown on the unit and intermediate packages.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.

6. NOTES

6.1 Ordering data.

6.1.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Type and shape required (see 1.2.1).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 National Stock Number (NSN) coverage. This specification covers the following items appearing in the Federal Supply Catalog:

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Stock No.	Item Identification	Specification Designation Type	Shape
6520-00-203-4620	WHEEL, ABRASIVE, Silicon Carbide, Straight Handpiece, Rounded Edge, 1/4 by 3/32 inches	II	A
6520-00-545-4960	WHEEL, ABRASIVE, Silicon Carbide, Angle Handpiece, Cone, 7/64 by 1/4 inches	I	D
6520-00-546-8000	WHEEL, ABRASIVE, Silicon Carbide, Straight Handpiece, Ball, 1/8 inch diameter	II	I
6520-00-547-0160	WHEEL, ABRASIVE, Silicon Carbide, Straight Handpiece, Cone, 7/64 by 1/4 inches	II	D
6520-00-547-5010	WHEEL, ABRASIVE, Silicon Carbide, Straight Handpiece, Cylinder, 3/32 by 7/32 inches	II	H
6520-00-547-6160	WHEEL, ABRASIVE, Silicon Carbide, Straight Handpiece, Inverted Cone, 3/16 by 5/64 inches	II	F
6520-00-547-7160	WHEEL, ABRASIVE, Silicon Carbide, Straight Handpiece, Knife Edge, 3/16 inch Diameter	II	C
6520-00-547-8010	WHEEL, ABRASIVE, Silicon Carbide. Straight Handpiece, Large Rounded Cone, 13/64 by 11/32 inches	II	E
6520-00-548-0010	WHEEL, ABRASIVE, Silicon Carbide, Straight Handpiece, Tapered Cylinder, 3/32 by 1/4 inches	II	G
6520-00-548-3010	WHEEL, ABRASIVE, Silicon Carbide, Straight Handpiece, Square Edge, 3/8 by 1/16 inches	II	B

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MILITARY INTERESTS

Custodians:

Army - MD
Navy - MS
Air Force-03

Review Activities:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

Project No. 6520-1820

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